

Microeconomic Analysis

Mark Dean

Homework 3

Due Thursday Oct 2nd

Question 1 Consider a binary relation $\succeq$ on some finite set X which is transitive and reflexive but not necessarily complete (which I will call an incomplete preference relation).

1. Show that there exists a utility function $u : X \rightarrow \mathbb{R}$ that represents $\succeq$ in the sense that

$$x \succeq y \rightarrow u(x) \geq u(y)$$

$$x \succ y \rightarrow u(x) > u(y)$$

2. Clearly this representation is ‘worse’ than the standard one, in the sense that we cannot recover the preference relation from the utility function. To get round this problem, we can use a **multi-utility representation**. A multi-utility representation of the relation $\succeq$ on X is a set of functions $\mathcal{U}$, where each $u \in \mathcal{U}$ is a function $u : X \rightarrow \mathbb{R}$, and these functions represent $\succeq$ in the sense that⁶

$$x \succeq y \text{ if and only if } u(x) \geq u(y) \forall u \in \mathcal{U}$$

Show that a multi utility representation has the same information as the original incomplete preference relation -i.e. given any multi-utility representation, there is a unique (incomplete) preference relation that is consistent with it"

3. Show that any transitive, reflexive relation on a finite set X admits a multi utility representation

Question 2 Status quo bias is the idea that choices can be influenced by the existence of a default, or status quo option. Consider the following model of status quo bias, which requires us to

extend our data. Let X be a finite set of alternatives, and $\mathcal{X}$ be all non empty subsets of X . We say that a pair $\{A, s\}$ is a choice problem if $A \in \mathcal{X}$ and either $s \in A$ or $s = \diamond$. We interpret $\{A, s\}$ as the choice from A when the status quo is s , with $s = \diamond$ indicating there is no status quo. Let $\mathcal{D}$ be the set of choice problems. We are going to place axioms on a choice correspondence C on $\mathcal{D}$ that allows us to characterize the following model: There exists

- $u : X \rightarrow \mathbb{R}^N$
 - A strictly increasing function $f : u(X) \rightarrow \mathbb{R}$
- such that

$$\begin{aligned} C(A, \diamond) &= \arg \max_{x \in A} f(u(x)) \\ C(A, x) &= x \text{ if } U_u(A, x) \text{ is empty} \\ &= \arg \max_{x \in U_u(A, x)} f(u(x)) \text{ otherwise} \end{aligned}$$

- Where $U_u(A, x) = \{y \in A \mid u_n(y) \geq u_n(x) \ \forall n \text{ and } u_n(y) > u_n(x) \text{ for some } n\}$

The interpretation of this model being that the DM ranks each option on a number of characteristics (through u). When faced with a choice problem with a status quo, they first check to see if there is anything that dominates the status quo, in the sense of being better along all dimensions. If not then they pick the status quo. If so, they think harder about all the options that dominate the status quo, aggregate the dimensions using f , and pick the best. If no status quo exists then they move straight to the ‘thinking harder’ stage.

1. Using the results from Question 1, show that this model is equivalent to the following: There exists

- A transitive, reflexive antisymmetric relation $\succeq$ on X
 - A completion $\triangleright$
- such that

$$\begin{aligned} C(A, \diamond) &= \{x \in A \mid x \succeq y \ \forall y \in A\} \\ C(A, x) &= x \text{ if } \nexists y \in A \text{ s.t. } y \succ x \\ &= \{y \in A \mid y \succeq z \ \forall z \succ x\} \text{ otherwise} \end{aligned}$$

2. We will now introduce our first two axioms;

Axiom 1 (Status Quo Conditional Consistency) *For any $s \in X \cup \diamond$, $C(., s)$ obeys axioms α and β (i.e. restricting to choice problems with the same status quo, the standard axioms hold)*

Axiom 2 (Status Quo Bias) *If $y \in C(A, x)$, then $y = C(A, y)$*

Define the binary relation $\succeq$ as

$$x \succeq y \text{ if } x \in C(\{x, y\}, y)$$

Show that if Status Quo Conditional Consistency and Status Quo Bias hold, then $\succeq$ is transitive, antisymmetric and reflexive.

3. Show that, under the same two axioms, for any choice with a status quo $x \in X$

$$C(A, x) \subseteq \begin{cases} x \text{ if } \nexists y \in A \text{ s.t. } y \succ x \\ \{y \in A | y \succ x\} \end{cases}$$

4. Now add two new axioms :

Axiom 3 (Dominance) *If $y = C(A, x)$ for some $A \subset B$ and $y \in C(B, \diamond)$ then $y \in C(B, x)$*

Axiom 4 (Status quo Irrelevance) *If $y \in C(A, x)$ and for every $\{x\} \neq T \subset A$, $x \notin C(T, x)$ then $y \in C(A, \diamond)$*

- (a) Show that, for any choice problem $\{A, x\}$ for $x \in A$, if $\{z \in A | z \succ x\}$ is non empty, then Status Quo Irrelevance implies that if $y \in C(A, x)$, then $y \in C(\{z \in A | z \succ x\}, \diamond)$
 - (b) Show that, for any choice problem $\{A, x\}$ for $x \in A$, if $\{z \in A | z \succ x\}$ is non empty, then Dominance implies that if $y \in C(\{z \in A | z \succ x\}, \diamond)$ then $y \in C(A, x)$
5. Use the above ingredients to show that a data set that is consistent with these axioms can be represented using either of the two models described above
6. Show that this model cannot explain the choice overload or choice difficulty effects we discussed in class

Question 3 Let $\succeq$ be a continuous and strictly monotonic preference relation on $\mathbb{R}_+^N$. Define

$$u(x) = \sup [\alpha \geq 0 | x \succeq (\alpha, \alpha, \dots, \alpha)].$$

1. Show that

$$x \sim (u(x), u(x), u(x) \dots u(x))$$

2. Show that u represents $\succeq$.

Question 4 Here are some questions involving classic results in random utility

1. Show that, if $|X| = 3$, then a stochastic choice data set has a random utility representation if and only if it satisfies monotonicity
2. Show that the Block Marshack inequalities imply monotonicity
3. To get some further intuition of what the Block Marshack inequalities are doing, consider the following. Let $X = \{x, y, z, w\}$. Let E_t be the event that a preference ordering is drawn such that $t \succ x$. Noting that

$$p(x, \{x, y, z, w\}) = 1 - P(E_y \cup E_z \cup E_w)$$

and using the classic inclusion/exclusion restrictions¹

$$\begin{aligned} P(E_y \cup E_z \cup E_w) &= P(E_y) + P(E_z) + P(E_w) \\ &- P(E_y \cap E_z) - P(E_y \cap E_w) - P(E_z \cap E_w) \\ &+ P(E_y \cap E_z \cap E_w) \end{aligned}$$

show that $p(x, \{x, y\})$ must satisfy the BM inequalities

4. Establish in the setting above that the BM inequality associated with $p(x, \{x, y\})$ is equal to the probability that $x \succ y$, $z \succ x$ and $w \succ x$.
5. Show the following three statements are equivalent

- (a) A random choice rule satisfies stochastic independence of irrelevant alternatives and positivity (i.e. $p(x, A) > 0$ for all $x \in A$)

¹Note that these can be derived from the standard inclusion exclusion restrictions by noting that

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

- (b) A random choice rule can be represented by the Luce model
- (c) There exists a utility function $v : X \rightarrow \mathbb{R}$ such that $u(x) = v(x) + \varepsilon_x$ for ε_x distributed according to the type 1 extreme value distribution and ε 's independent between any x and y in X . The probability of choosing x from A is then $P(u(x) > u(y) \ \forall \ y \in A)$

You can use the fact that, for any two independent extreme value random variables ε_x and ε_y the variable $\varepsilon = \varepsilon_x - \varepsilon_y$ has the CDF

$$F(\varepsilon) = \frac{e^\varepsilon}{1 + e^\varepsilon}$$